Team Number:	apmcm24209149
Problem Chosen:	A

2024 APMCM summary sheet

Begin to solve the two aspects of the problem, analyze the image and its classification, image analysis from the color, brightness, blurring to start, calculate the various aspects of the indicators, and intuitive visualization; and then establish a classification model, from the beginning of the threshold method to improve the normalization of the quantitative indicators of the weighted calculation of the normalization process, based on the final score as the basis for classification, the output results.

Question two is to establish an underwater degradation model after understanding the underwater imaging model provided in the question, analyze the images of different scenes, through mathematical methods to change the parameters of a certain aspect of the picture, and at the same time, take into account the influence of the attenuation of the light and the dimensions of the depth in the underwater scene, and establish the degradation model to analyze the similarities and differences of the different scenarios.

After that, based on the previous foundation, underwater enhancement model is established for a single scene, and picture enhancement is carried out by physical algorithms, and only one category of enhancement algorithm is used for a single scene, and the index is quantified to evaluate the enhancement results.

Considering the complex scenarios in the real situation, one method is to use multiple categories of enhancement methods comprehensively from the weighted scores of two indicators, UCIQE and UIQM, to come up with the final physical algorithm and to generate the images; another deep learning model is based on the U-Net architecture, and considering the lack of a large number of training sets and the ability to only carry out the task of training a small batch of samples, choosing this kind of convolutional network allows the use of fewer training images but still produce accurate image processing.

Compare various enhancement techniques with a single enhancement technique for complex scenes, compare the quantitative indexes of different methods, analyze the effect of enhancement, and visualize it, also think about the shortcomings, optimize the model, and put forward feasible suggestions for underwater visual enhancement from theory to practice. Reflect the significance of this technology.

Keywords: The normalization process   Enhancement algorithm   Deep learning model

Contents

I. Introduction

In order to indicate the origin of problems, the following background is worth mentioning.

1.1 Background

Underwater image enhancement technology is an important research direction in the field of computer vision, especially playing an important role in applications such as marine ecological research, underwater photography, ocean detection, and robot vision.

The propagation characteristics of light in the underwater environment are significantly different from those in the air, and the absorption and scattering of light by water can lead to image color distortion and contrast reduction. Molecules, particles and suspended matter in water absorb and scatter light, making underwater images generally lack rich colors and details, and image noise increases, further affecting image quality.

Numerous underwater image enhancement algorithms have been proposed in the last few years. However, these algorithms are mainly evaluated using either synthetic datasets or few selected real-world images.[2] So, we're going to come up with innovative approaches.

Underwater image enhancement techniques can help improve image quality and are more suitable for subsequent tasks such as visual analysis and target detection. Physical model-based enhancement methods mainly rely on the physical process of underwater image formation, which involves optical challenges inherent in underwater environments, such as light propagation, absorption, and scattering. Deep learning methods automate image enhancement by training neural networks to learn recovery rules from a large number of underwater images.

With the development of physical modeling and deep learning techniques, underwater image enhancement has become a multidisciplinary research field with important results in various practical applications and a need for future research.

1.2 Our Work

To avoid complicated description , intuitively reflect our work process, the flow chart is show as the following Figure 2

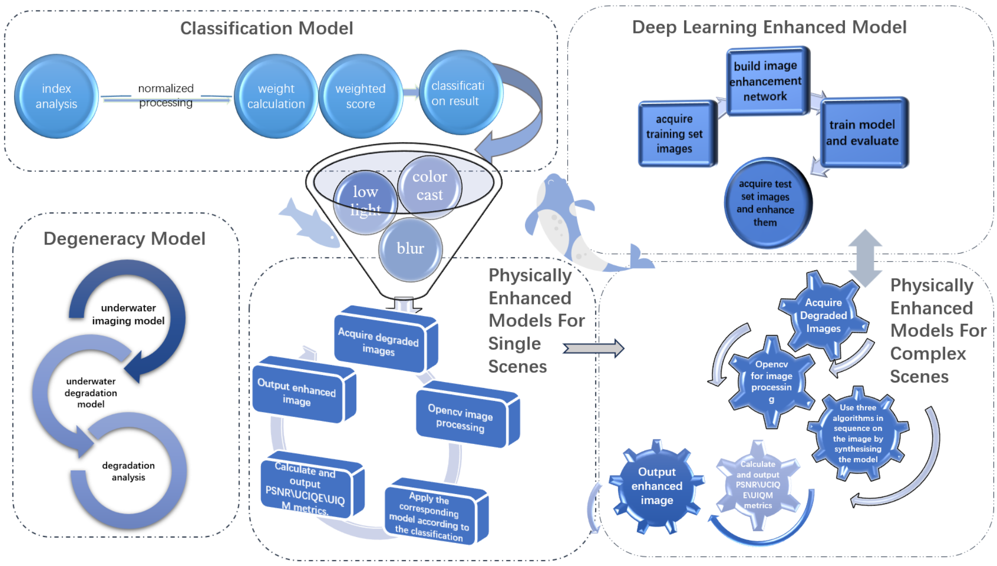

figure2:Flow chart of our work

First, we establish a classification model to analyze images from multiple angles for classification, followed by a degradation model to rationalize the degradation and analyze the similarities and differences.

Second, we establish a physical method of underwater image enhancement for a single scene based on the classification results, and quantify the index.

Third, the complex scene customizes the underwater image enhancement model, and builds physical and deep learning models by considering various factors.

Fourth, we calculate the metrics, evaluate the performance of our model in each environmental condition, and compare multiple methods to optimize the model.

Fifth, we summarize the results, conduct further exploration, and propose feasible recommendations.

II. The Description of the Problem

Underwater image enhancement technology supports marine resource exploration and environmental protection, underpins research in ocean engineering and marine biology, and drives the development of novel visual computing and artificial intelligence technologies. Our goal is to assess the degree of degradation of underwater images in different scenarios and provide targeted enhancement methods. Problem to be solved:

- Statistical analysis of images from multiple perspectives to determine the type of image and explain the classification.
- Construct an image degradation model for underwater scenes based on an underwater imaging model, assess the degree of degradation, and analyze the similarities and differences across scenes.
- Design and implement corresponding image enhancement methods for a single degradation type, validate the effectiveness of the methods, and quantitatively evaluate them.
- Considering multiple degradation characteristics comprehensively, build physical and deep learning underwater image enhancement models, applicable to complex scenes, and calculate quantitative indexes.
- Compare and optimize different image enhancement models on the basis of experimental validation, consider multiple factors comprehensively, and make feasible suggestions in practical applications.

III. Models

3.1 Classification Model

3.1.1 *Terms, Definitions and Symbols*

3.1.2 *Modelling and Solving*

It provides a comprehensive assessment of image quality by extracting a variety of image features (e.g. colour, brightness, blurriness, spectral energy, etc.). It is able to provide detailed degradation metrics for image quality analysis, and after establishing a classification model, classification is performed based on these metrics, and finally all results are exported to an Excel file for documentation and analysis.

Analysis of indicators :

Nomenclature / Notation	Define
N	Total number of pixels in the image
color_cast	Maximum difference between channels and threshold
mu_H	Tone Channel Mean
sigma_H	Standard deviation of colour channels
h_i	The hue value of the ith pixel in the image
mean_brightness	average brightness
Variance	the Laplace transform variance
L_i	Value of the ith pixel
magnitude	Fourier transform results
EH	Energy in the high frequency region
Blur	Difference between Laplace variance and threshold T_BLUR
low_light	Luminance Mean and Threshold Difference
g_i	The grey value of the ith pixel in the image

Table 1

1) Colour Shift Indicator:

The detection of colour shifts is mainly based on the RGB colour and HSV colour space analysis of the image:

After converting the image to RGB mode, the RGB is extracted and the mean value of the three channels is calculated, and the difference between the maximum difference between the colour channels and the threshold color _ cast with be used as an index.

$$\text{rgb\_diff} = \max\left(|\text{mean\_r} - \text{mean\_g}|, |\text{mean\_g} - \text{mean\_b}|, |\text{mean\_b} - \text{mean\_r}|\right)$$

$$\text{color\_cast} = |\text{rgb\_diff} - T_{\text{COLOR}}|$$

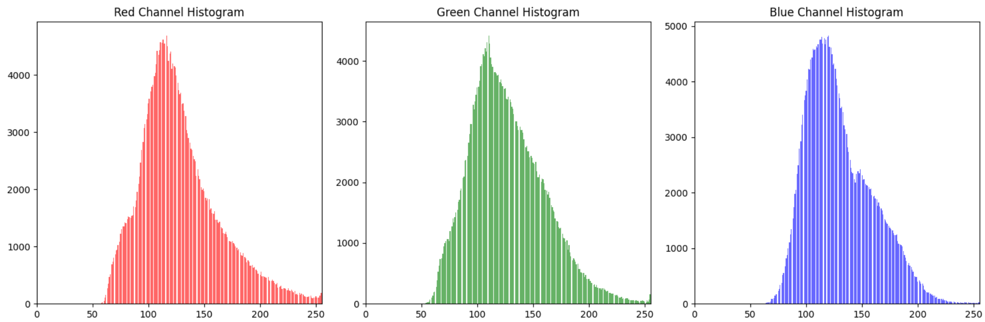

figure3:Single Channel Histogram

As shown in the figure3:

x-axis: pixel intensity values (0 to 255), indicating the intensity of the colour channel.

y-axis: frequency, indicating the number of times a certain intensity value appears in the image.

The histogram visualises the colour distribution of the tested image on the red, green and blue channels respectively.

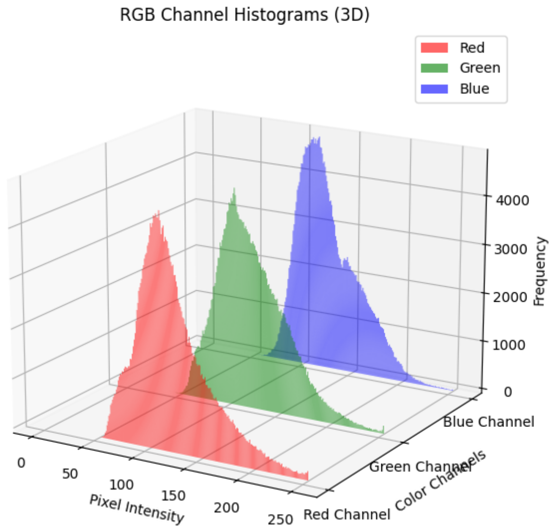

figure4:RGB Histogram

As shown in the figure4:

x-axis: pixel intensity values (0 to 255), indicating the intensity of the colour channel.

y-axis: frequency, indicating the number of times a certain intensity value appears in the image.

The histogram visualises the comparison of the colour distribution of the tested images.

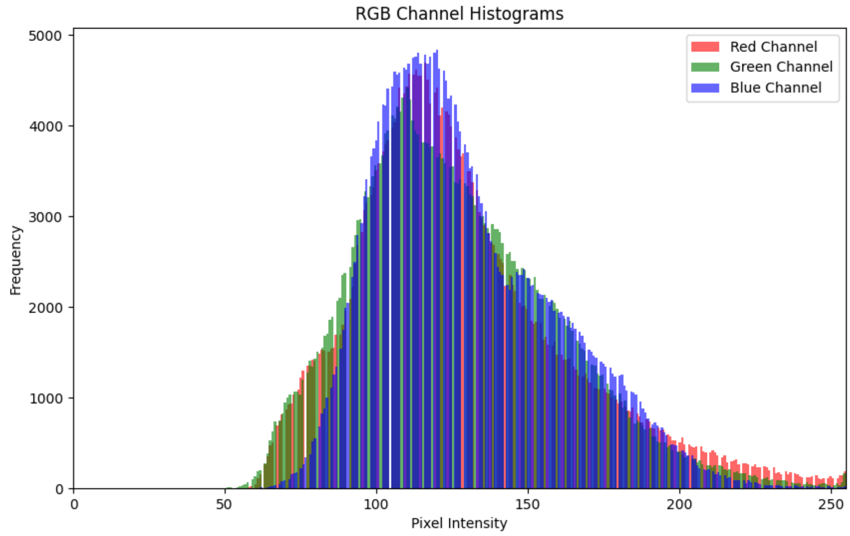

figure5:RGB3D Graph

As shown in the figure5:

X-axis: Pixel Intensity, i.e. pixel value, ranging from 0 to 255.

Y-axis: Colour channel

Z-axis: Frequency, i.e. the number of times each pixel intensity value (each value on the X-axis) appears in the image.

Look at the pixel intensity distribution of the three colour channels and their frequencies at the same time to visually compare the pixel distribution of each colour channel.

After converting the input image from BGR colour space to HSV colour space, extract the hue (Hue) channel in the HSV image and calculate the mean (mu_H) and standard deviation (sigma_H) of the hue channel.

The formula for calculating the mean value (mu_H):

$$u_H = \frac{1}{N}\sum_{i=1}^{N} h_i$$

Standard deviation (sigma_H) Calculation formula:

$$\sigma_H = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(h_i - H_H)^2}$$

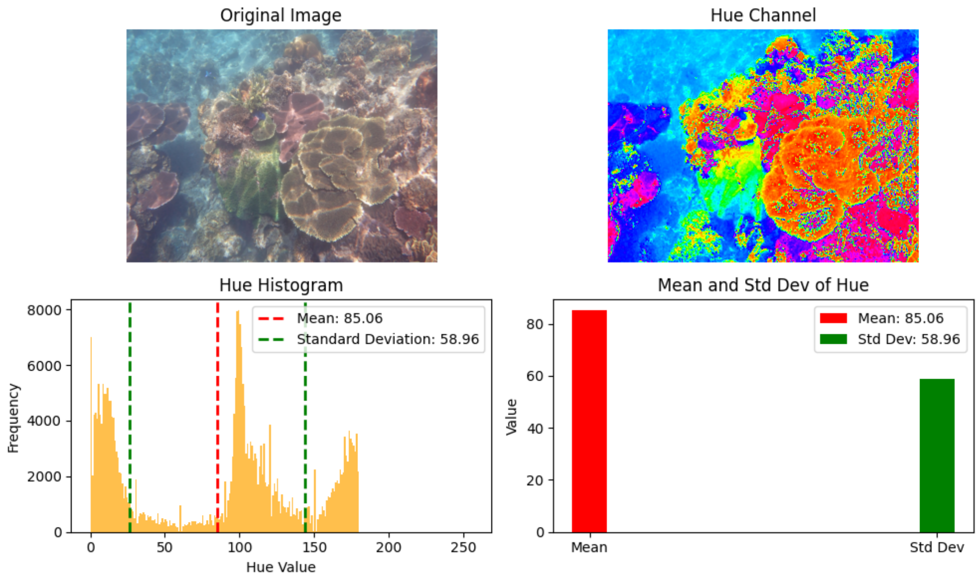

figure6:Luminance Chart

As shown in figure6:

Figure 4, top left, shows the original image;

Figure 4 top right shows the display hue (Hue) channel;

Different hue values are rendered in different colours, and with this figure you can see how the hue values are distributed for each pixel in the image.

Figure 4, bottom left, plots the hue histogram X-axis (Hue Value): represents the value of the hue, the range is set to 0 to 255 each bar corresponds to an interval of the hue value, the value on the X-axis is the intensity of the different hues.

Y-axis (Frequency): represents the frequency of each hue value in the image, i.e. how many pixels have a hue value falling within this interval.

The histogram shows the distribution of all pixel values in the hue channel, while the mean and standard deviation of the hues are marked by vertical dashed lines.

Figure 4, bottom left, shows the mean and standard deviation of the hue.

The X-axis has only two scales, Mean and Std Dev, which are distinguished by [0] and [1].

Y-axis (Value): indicates the specific value of the Mean and Std Dev.

The red bar represents the mean value of the hue, giving where the hue is concentrated. The green bar represents the standard deviation of the tones and shows the width of the distribution of the tone values. When the standard deviation is large, there is a wide variety of colours in the image; when the standard deviation is small, the hues in the image may be concentrated in a certain range.

2) Low light indicator:

Converts an image to a greyscale image and calculates the average brightness and the standard deviation of the brightness for that greyscale image, A greyscale image is a single-channel image in which each pixel has a value that indicates the brightness of that pixel (from black to white). After conversion to greyscale image, each pixel of the image has only one value indicating the brightness. The standard deviation of the entire greyscale image is calculated, indicating the degree of dispersion of the image's luminance values.

Mean_brightness (mean_brightness) formula:

$$Mean\_\_brightenss = \frac{1}{N}\sum_{i=1}^{N} g_i$$

Standard deviation (std_dev_brightness) formula:

$$Std\_\_dev\_\_brightness = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(g_i - u)^2}$$

Mean and Threshold Difference as an Indicator:

$$low\_\_light = abs(mean\_\_brightness - T\_\_LIGHT)$$

figure7:Grey Scale

As shown in Figure 7:

Shows the greyscale map, the average luminance, and the standard deviation of the luminance.

3) Fuzzy indicator:

Calculate the Laplace transform of an image and compute the variance. The Laplace transform calculates the second-order spatial derivative of an image and is typically used to detect high-frequency portions of an image (e.g., edges). The stronger the Laplace transform of an image, the clearer the information about the edges of the image and the sharper the image; conversely, the blurrier the image.

Laplace Transform Variance:Variance is a statistic of the Laplace Transform result and is used to measure the degree of blurring in an image.

Variance formula:

$$Variance = \frac{1}{N}\sum_{i=1}^{N}(L_i - u)^2$$

Calculate the difference between the variance of the image and the threshold T BLUR as an indicator of the blurriness of the image.

$$blue = |variance - T_{BLUR}|$$

Calculate the two-dimensional Fourier transform of an image. The Fourier transform transforms the image from the spatial domain to the frequency domain, where the frequency information describes the variations at different spatial scales in the image. The zero-frequency portion of the spectrogram (i.e. the DC component) is moved to the centre of the spectrum to make it easier to view the frequency distribution. Typically, the zero-frequency portion is located in the upper left corner of the spectrum Move it to the centre of the image. Calculate the magnitude spectrum of the Fourier transform result. The output of the Fourier transform is a complex number and the magnitude spectrum is the mode of the complex number, the formula i.e.:

$$Magnitude = \sqrt{Real(fshift)^2 + I_{mag}(fshift)^2}$$

Extracting the high-frequency portion of the spectrum, usually after the Fourier transform, the high-frequency components of the image (details and edges) are located in the upper-right corner of the spectrum Calculate the energy of the high-frequency region. The high frequency region contains the details and edges of the image and the energy is the sum of the squares of the frequency components of the region.

The equation for the energy of the high frequency region:

$$E_H = \sum_{i,j} |high\_freq_{i,j}|^2$$

EH: Energy in the high frequency region as an indicator.

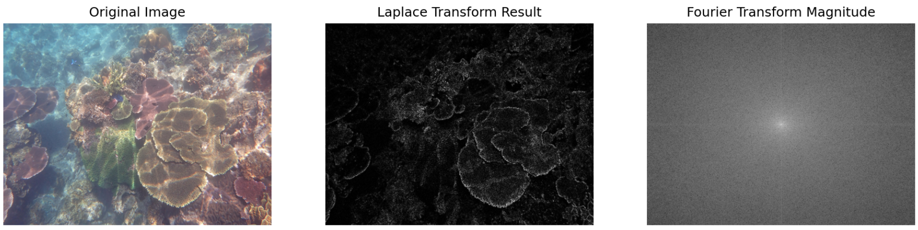

figure8:Fuzzy Indicator Chart

As shown in Figure 8:

Figure 6 left demonstrates the visual information of the original image.

Figure 6 centre shows Laplace Transform results - Demonstrates the results of edge detection after Laplace Transform, highlighting the edges and fast changing regions in the image.

Figure 6 right shows Fourier Transform Magnitude - demonstrates the frequency domain representation of the image, looking at the frequency components of the image, the distribution of low and high frequency information through the Fourier Transform.

Define the weights for each indicator :

- Prioritise the more critical metrics:

  Colour Cast and Low Light: These are major factors in image quality. For example, an image with a high colour cast can hardly be considered 'clear' or 'natural', while an image in low light conditions can appear blurred and difficult to recognise. Therefore, these two metrics are of higher importance and are weighted accordingly.

  Blur: Whether or not an image is blurred also has a significant impact on visual quality, but blurring may be temporary or related to other factors (e.g. low light). It is therefore given a medium weighting.

- The role of supporting indicators:

  Tone Mean (mu_H) and Tone Standard Deviation (sigma_H): These metrics are used more for analysing the colour distribution of an image, and may be critical for some applications (e.g., automatic colour adjustment, white balance correction), but may be relatively unimportant in other applications. Therefore they are given a relatively low weight.

  Standard deviation of brightness (std_dev_brightness) and Energy of High

Frequency (EH): these metrics reflect the brightness variation and spectral characteristics of an image. In some cases, they have an impact on the sharpness of the image, but may play a smaller role compared to the impact of colour and low light, so their weights are set low.

Normalisation Next, the indicators are normalised to scale their values to the range [0, 1]. The purpose of normalisation is to remove the differences in the scales of the indicators so that they can fairly participate in the weighted summation.

The normalisation formula is:

$$Normalizae\_\_value = \min(\frac{raw\_\_value}{\max\_\_value}, 1)$$

image file name	normalized_color_cast	normalized_low_light	normalized_blur	normalized_mu_H	normalized_sigma_H	normalized_std_dev_brightness	normalized_EH	score
image_001.png	1	1	1	0.273267353	0.022840782	0.724717835	0.019620377	0.781472111
image_002.png	0.104147859	1	0.770216772	0.178857389	0.073272439	0.91243317	0.009038921	0.510761722
image_003.png	0.680519051	0.732145433	0.611865371	0.230552175	0.072065455	1	0.059159419	0.573244509
image_004.png	0.516746701	0.154815885	0.948592117	0.211371118	0.149560025	0.441817594	0.00267392	0.426629012
image_005.png	1	1	0.818784409	0.282294937	0.06150854	0.498973994	0.015715033	0.738772179
image_006.png	1	1	0.675344617	0.197962071	0.052773819	0.684220491	0.00304839	0.70455648
image_007.png	1	0.770082457	0.927044767	0.280758094	0.022299702	0.635254011	0.001597357	0.704195688
image_008.png	1	0.868983986	0.724612755	0.27815028	0.026438359	0.564037592	0.036535305	0.688390335
image_009.png	1	0.832911774	0.877530109	0.281556586	0.068020066	0.727814474	0.001162601	0.719276444
image_010.png	0.130357297	1	0.847023808	0.250870788	0.054134976	0.503971419	0.53673363	0.573910136
image_011.png	1	0.737995988	0.876380544	0.260151442	0.027561308	1	0.352564722	0.736810425
image_012.png	1	0.598196586	0.884163933	0.292445207	0.051754572	0.688101398	0.064867351	0.666315976
image_013.png	0.685227598	1	0.115140659	0.246167402	0.07178166	0.820570778	0.037928787	0.533259725
image_014.png	0.908235547	1	0.594660344	0.201089328	0.554065816	0.665546588	0.005565267	0.715394793
image_015.png	0.502194401	1	0.3181399	0.253981554	0.378608876	0.559157659	0.005276622	0.543620246
image_016.png	1	0.851402863	0.852087308	0.222939276	0.227904612	0.819491225	0.113369416	0.741811033
image_017.png	0.15345625	1	0.565839836	0.277526015	0.231226455	0.477387651	0.003203099	0.49047327
image_018.png	0.506259766	0.066339062	0.679012909	0.261310505	0.089726131	0.425389999	0.02556091	0.350947069
image_019.png	0.611255273	0.38603444	0.993554626	0.286927816	0.053674558	0.147738191	0.001942538	0.504021145
image_020.png	0.914625391	0.493594748	0.95289481	0.27708648	0.415791745	0.676248185	0.05751183	0.665339736
image_021.png	0.837953798	0.171172721	0.885790556	0.217597918	0.37452793	0.768547445	0.070955758	0.54505517
image_022.png	0.265395833	1	0.518016148	0.276912851	0.248742291	0.450648331	0.003432613	0.509239023
image_023.png	0.917244206	0.162636827	0.626839573	0.213359429	0.499138994	0.892025497	0.111735357	0.533030797
image_024.png	0.817452387	0.295311306	0.878500185	0.137903025	0.345128431	0.663934977	0.076526347	0.549938641
image_025.png	0.837863911	0.322908984	0.968854644	0.17605775	0.380504056	0.715915573	0.061789166	0.590397916

figure9:Metrics Calculation Chart

As shown in figure 9:

The figure shows the specific values calculated for the indicator after normalisation of the test charts in the Annex section.

Calculating the weighted score :

In this section, the composite score of the image is calculated by means of a weighted sum. The values of the individual normalised metrics are multiplied by their weights and all the weighted values are added to get a total score. This total score reflects the quality of the image.

The formula for calculating the weighted score is:

$$Score = \sum_i (normalized\_\_value_i \times weight_i)$$

Classification according to score :

Based on the weighted score calculated, the programme will automatically classify the images into different types and export the results.

3.1.3 *Analysis of the Result*

Pros :

- Flexibility and adjustability of the model: By setting weights for different metrics, we can adjust their contribution to the classification results according to the actual situation. For example, if an application scenario is particularly concerned with the colour accuracy of an image and less concerned with brightness or sharpness, the weights can be adjusted to make the impact of colour bias greater.
- Simplify and avoid complexity: By setting reasonable weights, overly complex calculations or judgements for each indicator are avoided. After setting the weights, the method of calculating the score becomes more direct and easy to understand, and excessive conditional branching is avoided, making the code concise and efficient. Normalising each metric allows metrics with different scales to be compared under the same criteria. This avoids the unbalanced impact of differences in different units of measurement (e.g. brightness and hue) on the final score.
- Multi-dimensional comprehensive assessment: The method does not only focus on a single metric (e.g., blur or low light), but assesses image quality comprehensively through multiple metrics. Each indicator reflects a different quality problem of the image, and the combined consideration of these indicators can provide more comprehensive and accurate image classification results. Multi-dimensional assessment reflects the actual performance of the image in a more comprehensive way, and is applicable to the comprehensive judgement of image quality.

Disadvantages :

- Static classification threshold: For different application scenarios, the classification threshold may need to be adjusted according to the actual situation, otherwise it may lead to errors in classification. Lack of dynamic adjustment mechanism. If the overall level of image quality changes, these thresholds may need to be reset.
- Sensitivity to extreme values: If the value of a metric is too high or too low, the normalisation may limit its value to 1 (maximum). This means that extreme values may not truly reflect their impact on image quality when calculating the weighted score, affecting the accuracy of the classification.

3.2 Degeneracy Model

3.2.1 *Terms, Definitions and Symbols*

Nomenclature / Notation	Define
I(x)	Degraded underwater image
J(x)	Clear image
β	Ambient light in an underwater environment
t(x)	Transmission function for an underwater scene
T	The degree to which light is attenuated through the medium
β(beta	Indicates the attenuation coefficient for each color channel
δ(depth)	The depth value of each pixel
Jc	Color component of the original image
Tc	Transmission coefficients for each color channel
I(x,y)	Pixel values of the image
G(i,j)	Gaussian kernel

Table 2

3.2.2 *Modelling and Solving*

The Jaffe-McGlamery underwater imaging model is a model used to characterize light propagation in underwater imaging systems. It is mainly used to explain the degradation process of underwater image quality, especially the effect of light attenuation and scattering on image quality in environments with large water depths or poor water quality. This model is commonly used in underwater photography, oceanographic research, environmental monitoring and other fields.For additive noise, we demonstrate that the nonlinear NQM is a better measure of visual quality than peak signal-to noise ratio (PSNR) and linear quality measures.[1]

In general, an embedded imaging model is used due to the close proximity between the object and the camera:

$$I(x) = J(x)t(x) + B(t(x))$$

The degradation model is based on this formula for parameter tuning to establish the degradation model.

figure10:Degraded model comparison

As shown in the figure10, the original image and the three degraded images are compared, and the difference between the degraded processing and the original image can be visualized.

The Transmission Coefficient formula calculates the light transmission coefficient for each pixel of an image, which is related to the depth of the object (depth) and the attenuation coefficient (beta), which indicates how much light attenuates as the depth of the object increases. In the calculation, larger depth values (objects farther away) result in higher attenuation.

$$T(d) = e^{-\beta d}$$

Color offset degradation model :

The color offset degradation of the simulated image takes into account the depth information and the different attenuation rates of each color channel, and the degraded color components are calculated for each color channel (red, green and blue) according to the following equation:

$$I_c = J_c T_c + B_c(1 - T_c)$$

By calculating the transmission coefficients of each color channel and then weighting and averaging the clear image and the background image to generate the degraded image, in this way, the color changes at different depths can be simulated, so as to achieve the effect of simulating haze and atmospheric attenuation.

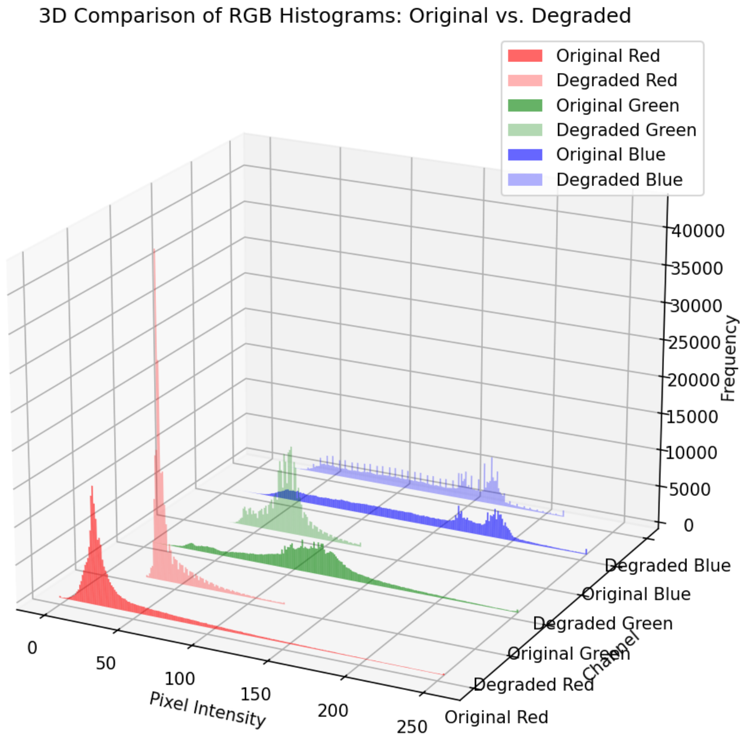

figure11:RGB Degradation Histogram

As shown in the figure11, X-axis: pixel intensity; Y-axis: color channel; Z-axis: frequency, a 3D histogram is plotted to compare the pixel distribution of the original and color-shifted images on the RGB channel.

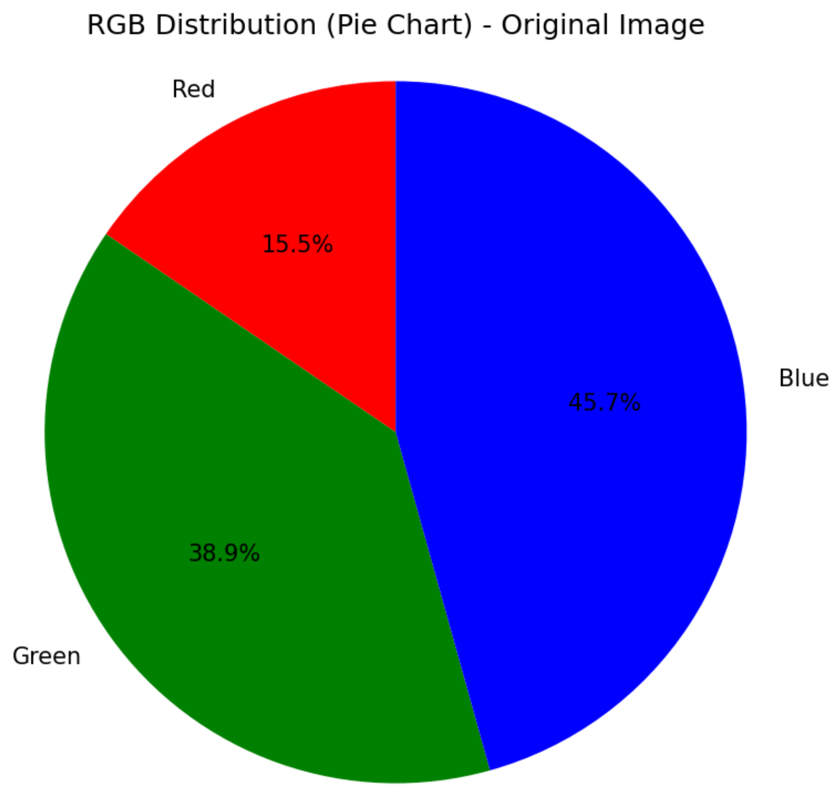

figure12:RGB Pie Chart

As shown in the figure12, the three RGB color channels are extracted and the sum of each channel is calculated as a percentage of the total number of pixels in the image, and a pie chart representing the color distribution of the image is drawn. The output pie chart can visualize the proportion of different colors in the image and help analyze the color distribution of the image.

Low-light degradation model :

The degradation effect of the image under low light is simulated. The brightness of the image is adjusted by balancing the weight between the input image and the background image through the transmission coefficient T. As the depth increases, the brightness of the image gradually decreases and becomes more dependent on the background light, and the degraded image is calculated as:

$$I = JT + B(1 - T)$$

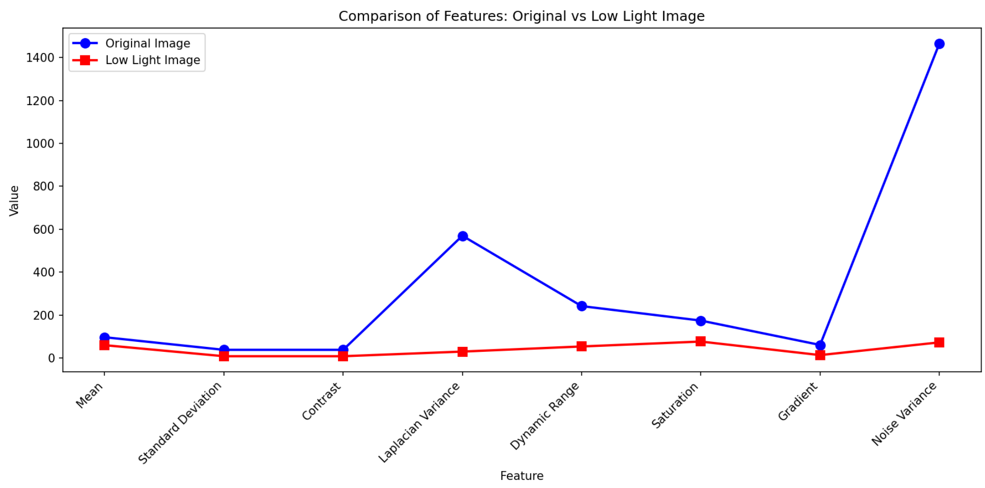

figure13:Feature Comparison Chart

As shown in the figure13:

Horizontal axis: name of the feature

Brightness Mean: converts the image to a grayscale map (i.e., removes the color information) and then calculates the mean of the brightness of each pixel.

Standard deviation: Used to measure how much the brightness of pixels in an image fluctuates.

Laplace operator (local contrast and sharpness):Used to detect edges and details in an image, and can therefore be used to assess the sharpness of an image. The Laplacian variance of an image reflects how much detail is in the image.

$$LaplacianVariance = Var(\nabla^2 I)$$

Dynamic Range:The range of luminance in an image, i.e. the difference between the maximum and minimum luminance values.

$$DynamicRange = I_{max} - I_{min}$$

Color Saturation:Measures how vivid the colors in an image are. A more saturated image has more vivid colors.

Image Gradient:Describes the rate of change of brightness in an image, usually used for edge detection, by calculating the Sobel gradient operator of an image (along the x and y directions, respectively), and then calculating the gradient magnitude at each pixel point.

$$Magnitude = \sqrt{(\frac{\partial I}{\partial x})^2 + (\frac{\partial I}{\partial y})^2}$$

Image noise: The noise is estimated by calculating the variance of the image brightness.

$$NoiseVariance = Var(I)$$

Vertical axis: the value of each feature, the larger the value, the more prominent the image features.

Blue line: the feature value of the original image.

Red line: the feature value of the low-light processed image.

Modeling of blurring effect To add blurring effect to the image, Gaussian blurring algorithm is used to smooth the image and the details are blurred.

Gaussian blur is a commonly used smoothing filter to reduce noise and details in an image. It performs a weighted average of pixels based on a normal distribution. The intensity of the blur can be controlled by adjusting the size and standard deviation of the convolution kernel.

Gaussian kernel calculation formula:

$$G(i, j) = (\frac{1}{2\pi\sigma^2})^{-\frac{i^2+j^2}{2\sigma^2}}$$

The formula for Gaussian fuzzy is:

$$I = \sum_{i=-k}^{k} \sum_{j=-k}^{k} I(x+i, y+j) \cdot G(i, j)$$

3.2.3 *Analysis of the Result*

Underwater images are affected by a variety of factors, including the propagation characteristics of light, the turbidity of the water body, and the angle of incidence of light. In different scenes (e.g. color projection, low light, etc.), the degradation of underwater images shows different characteristics. The following is an analysis of these degradation models, classified in terms of color, clarity, etc.

Degradation analysis :

- Color Projection Scene:
  - Color Distortion:

    Colors in underwater environments can be distorted due to light absorption and scattering, especially in deeper waters, where red and orange wavelengths of light decay rapidly, resulting in bluish or greenish underwater images.

  - Reduced Color Saturation:

    Due to the turbidity of the water and the scattering of light, the color saturation of an image may be reduced, resulting in an image that looks flat.

  - Reduced Contrast:

    Scattering of light can make an image less contrasty and blur details.

- low light scenes:

  - Low Brightness:

    In a low light environment, the brightness of the image is obviously insufficient, resulting in increased image noise.

  - Blurring:

    Due to the lack of light, the shutter speed may be reduced, resulting in blurred images.

  - Loss of details:

    Under low light conditions, details may be masked by noise, resulting in loss of important information.

- Other scenes (e.g. turbid waters, glare reflections, etc.):

  - Turbid waters:

    the turbidity of the water body will lead to increased light scattering and absorption, affecting the clarity and color of the image.

  - Strong Light Reflections:

    Under strong light conditions, reflections from the water can lead to overexposure of the image and loss of detail.

Analysis of Similarities and Differences :

- Similarities: Degradation in all scenes is affected by the optical properties of water, including scattering and absorption.

  Color distortion and contrast degradation are common to a wide range of scenes.

- Differences:

  Color cast scenes place more emphasis on color distortion and color saturation degradation, while low light scenes are more concerned with lack of brightness and noise effects.

  In turbid waters, changes in turbidity have a more significant effect on image quality, while in bright light reflections, reflection effects may lead to overexposure.

3.3 Physically Enhanced Models For Single Scenes

3.3.1 *Terms, Definitions and Symbols*

Nomenclature / Notation	Define
PSNR	The similarity of the processed image to the original image
UCIQE	Underwater image quality
UIQM	Color Evaluation Criteria
RGB color space	Color model
ϵ	Prevents the offset of dark area compression
Fourier Transform	Filtering operations
Wiener Filter	Methods for deblurring and noise
Richardson-Lucy algorithm	Iterative deambiguation algorithm
Guided Filter	Filtering algorithm that preserves edges

Table 3

3.3.2 *Modelling and Solving*

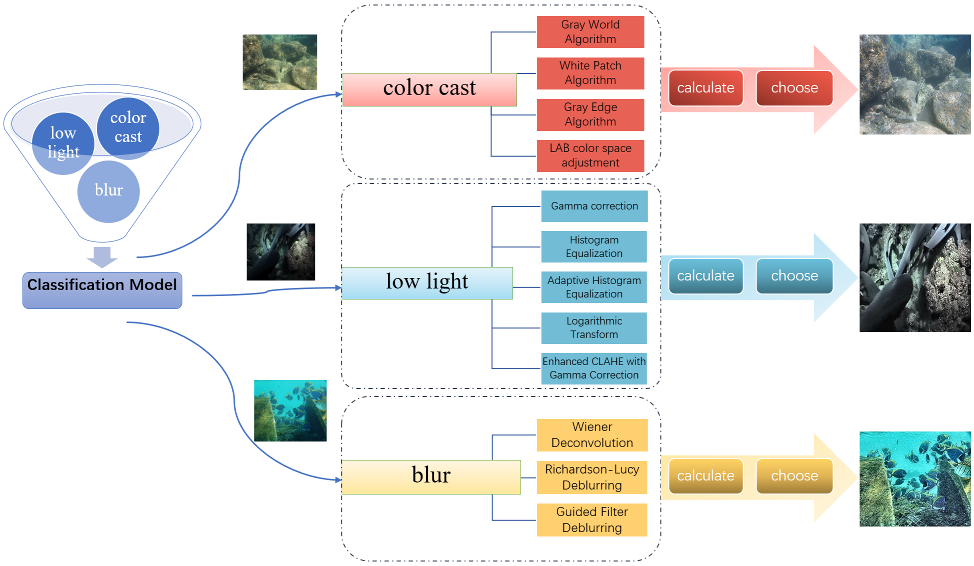

figure14:Flowchart of a physically enhanced model for a single scenario

Since there are more models for image enhancement using physical methods, the enhancement effect of different algorithms varies in different scenarios. Therefore, we use multiple algorithms to calculate simultaneously and select the best algorithm through the values of UCIQE and UIQM to achieve the best effect enhancement.

In each type, through the steps of image data type conversion, channel separation, statistical calculation, scaling adjustment and value limitation, and according to different image characteristics and index values, we choose the appropriate algorithm or use it in combination to achieve the best enhancement effect.

Eventually, combined with the classification model proposed in Problem 1, targeted enhancement of a single type of image is performed, and the enhanced image is output to a specified folder, while the PSNR UCIQE UIQM metrics are calculated and the metrics are output to an excel sheet.

Color Shift Enhancement :

For the color offset image, we use four algorithms for enhancement, namely, grayscale world assumption, white spot enhancement algorithm, grayscale edge algorithm, and LAB color space adjustment, and based on the selection of algorithms, the algorithm with the best enhancement effect is finally adopted for output.

Model building:

- Grayscale world assumption:

  Given a color image II (size M×N), each pixel of the image contains R,G,B three channel values. Due to lighting conditions (e.g., the color of the light source), the image may suffer from an overall color bias problem. The goal is to make the overall average color value of the image converge to grayscale (neutral color) by adjusting the values of each channel. Here, we assume that the image pixels are evenly distributed and the average value of all pixels can represent the overall color of the image.

Ideally, the average image value satisfies Rmean = Gmean = Bmean.

Formula:

Calculate the mean value of each channel:

$$\text{avg\_\_gray} = \frac{\text{avg}_R + \text{avg}_G + \text{avg}_B}{3}$$

Gain coefficient:

$$\text{scale}_R = \frac{\text{avg\_\_gray}}{\text{avg}_R}, \quad \text{scale}_G = \frac{\text{avg\_\_gray}}{\text{avg}_G}, \quad \text{scale}_B = \frac{\text{avg\_\_gray}}{\text{avg}_B}$$

Apply gain and merge channels:

$$\text{Balanced} = \text{merge}(R \times \text{scale}_R,\ G \times \text{scale}_G,\ B \times \text{scale}_B)$$

- White Spot Enhancement Algorithm:

  Given an input image I (size M×N), the brightest pixel in the image should be white (i.e., the RGB channel values are maximized) under ideal light conditions.

  The brightest pixel in the image should be white (i.e., the RGB channels have the largest values) under ideal lighting conditions. The White Spot Enhancement algorithm corrects the color deviation by normalizing the value of each channel to the maximum value of that channel.

  Calculate the high percentile (e.g. 0.995) for each channel:

$$\text{max}_R = \text{percentile}(R, 99.5), \quad \text{max}_G = \text{percentile}(G, 99.5), \quad \text{max}_B = \text{percentile}(B, 99.5)$$

Scaling factor:

$$\text{scale}_R = \frac{\text{max\_\_RGB}}{\text{max}_R}$$

,

$$\text{scale}_G = \frac{\text{max\_\_RGB}}{\text{max}_G}$$

,

$$\text{scale}_B = \frac{\text{max\_\_RGB}}{\text{max}_B}$$

where

$$(\text{max\_\_RGB} = \max(\max_R, \max_G, \max_B))$$

Apply scaling and merge channels:

$$\text{Balanced} = \text{merge}(B \times \text{scale}_B,\ G \times \text{scale}_G,\ R \times \text{scale}_R)$$

- Grayscale Edge Algorithm:

  We assume that the average color value of all edge regions in an image should be close to gray (i.e., R=G=B). the edge information is defined by the region with the most pronounced change in the gradient. Given a color image I, the goal is to adjust the three RGB channels according to their gradient information so that the color distribution of the edge regions is closer to neutral gray.

  Formula:

  Calculate the Laplacian gradient for each channel: [Gradunderline R = Laplacian(R), Gradunderline G = Laplacian(G), Gradunderline B = Laplacian(B)]

  Calculate the gradient mean:

  $$\text{avgGray} = \frac{\text{avgGrad}_R + \text{avgGrad}_G + \text{avgGrad}_B}{3}$$

  Scaling factor:

  $$\text{scale}_R = \frac{\text{avgGray}}{\text{avgGrad}_R}$$
  $$\text{scale}_G = \frac{\text{avgGray}}{\text{avgGrad}_G}$$
  $$\text{scale}_B = \frac{\text{avgGray}}{\text{avgGrad}_B}$$

  Apply scaling and merge channels:

  $$\text{Result} = \text{merge}(R \times \text{scale}_R\ G \times \text{scale}_G\ B \times \text{scale}_B\ R)$$

- LAB Color Space Adjustment Algorithm:

  The A and B channels of the LAB space represent the red, green, yellow and blue color information, respectively. Under the ideal white balance condition, the average value of A and B channels should be close to zero, indicating that the overall color neutrality of the image. Therefore, the core of the LAB white balance algorithm is to adjust the A and B channels so that their average value goes to zero.

  Formula:

Convert to LAB space:

$$\text{LAB} = \text{RGB2LAB}(Image)$$

Split channels:

$$L, A, B = \text{split}(\text{LAB})$$

Adjust A and B channels:

$$A' = A - (\text{avgA} - 128), \quad B' = B - (\text{avgB} - 128)$$

Limit range and merge:

$$A' = \text{clip}(A', 0, 255), \quad B' = \text{clip}(B', 0, 255)$$

Convert back to RGB space:

$$\text{Result} = \text{LAB2RGB}(\text{merge}(L, A', B'))$$

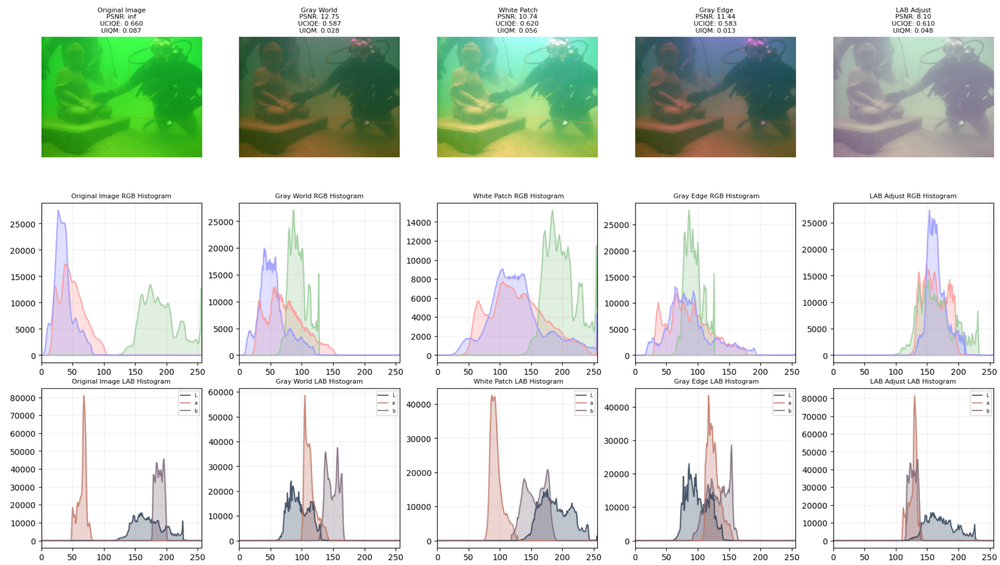

figure15:Histogram comparison

As shown in the figure15: First row: original and enhanced images with evaluation metrics.

The second row is an RGB histogram, with the horizontal coordinate indicating the intensity range of pixel values and the vertical coordinate indicating the number of pixels.

The third row is a LAB histogram with the horizontal coordinate indicating the channel luminance value and the vertical coordinate indicating the number of pixels.

From the histogram, it can be seen that each algorithm has a certain white balance effect on the original image. In this example, the White Patch algorithm has the highest weighted average of UCIQE and UIQM values, so the enhancement result of the White Patch algorithm is chosen as the final result.

Low light enhancement :

For color shifted images, we use five algorithms for enhancement: improved gamma transform, histogram equalization, improved adaptive histogram equalization, logarithmic transform enhancement, and improved CLAHE combined with gamma enhancement, and select the algorithms according to the weighted selection algorithms, and ultimately the algorithms with the highest scores are used for the output.

Modeling:

- Improved Gamma Transform: principle:

  Gamma transform is a nonlinear gray level mapping used to adjust the brightness of an image. By performing exponential operations on the luminance channel of an image, the brightness of an image can be enhanced or diminished to improve image visibility under low light conditions.

$$I_{\text{out}} = I_{\text{in}}^{\gamma}$$

  Equation

$$I_{\text{out}} = I_{\text{in}}^{\gamma}$$

  where:

  (I_{in}) is the input pixel value (normalized to the range [0, 1]).

  (γ) is the gamma value:

  ($\gamma < 1$) Enhances the brightness.

  ($\gamma > 1$) diminish brightness.

- Histogram equalization:

  Histogram equalization enhances the contrast of an image by adjusting the brightness distribution of the image to make the histogram of the image more uniform. This is especially effective in low-contrast images.

  The Formula

  The core of histogram equalization is the cumulative distribution function

(CDF). Assuming that the input image has a gray level of (I) and an output gray level of (I'), then:

$$I'(x, y) = \text{CDF}(I(x, y)) \times (L - 1)$$

Where:

$$\text{CDF}(I) = \sum_{j=0}^{I} P(j)$$

($P(j)$) is the probability of the gray level (j).

(L) is the total number of gray levels (usually 256).

- Improved Adaptive Histogram Equalization (AHE):

  Adaptive histogram equalization (ahe) is a contrast enhancement method designed to be broadly applicable and having demonstrated effectiveness. [3] Adaptive Histogram Equalization divides the image into multiple tiles and applies histogram equalization independently to each tile, avoiding the over-enhancement or noise amplification that may result from global histogram equalization. The enhancement is further controlled by the contrast limit parameter (clip limit).

  Formula

  CLAHE applies the following formulas to each of the clips:

$$I'\text{CLAHE}(x, y) = \text{CDFclip}(I(x, y)) \times (L - 1)$$

  where:

  (CDF__clip) is the cumulative distribution function after contrast limitation.

- Logarithmic Transform Enhancement (Logarithmic Transform):

  Logarithmic transformation enhances the dark details of an image by taking the logarithm of the image pixel values, which compresses the high gray level information and expands the low gray level information. This works well for brightening low light images.

  Equation.

$$I_{\text{out}} = c \cdot \log 1 + I_{\text{in}}$$

- Enhanced CLAHE with Gamma Correction:

  Principle

Combining CLAHE and Gamma Correction, local contrast is first enhanced by CLAHE, then gamma correction is applied to adjust the brightness, and finally the image quality is further improved by local contrast enhancement and detail layer extraction.

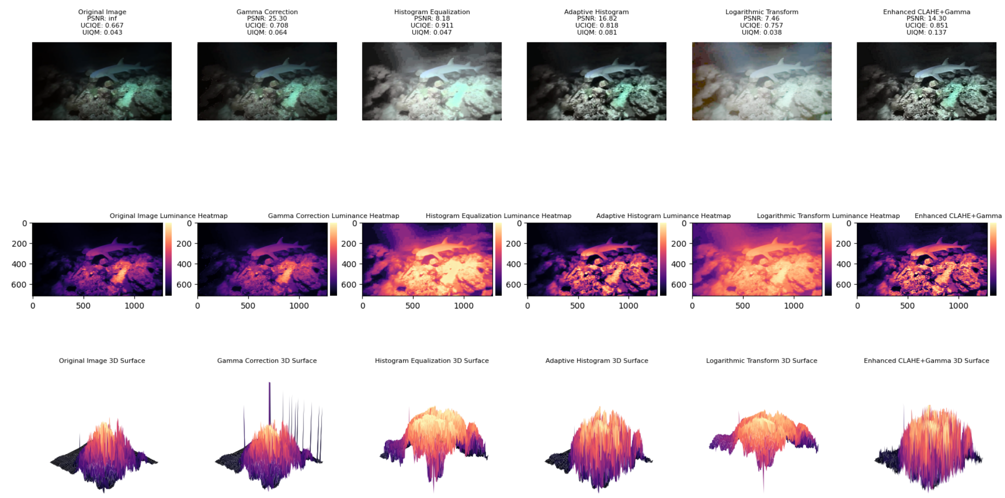

figure16:Different brightening algorithms enhance the picture

As shown in the figure16:

First row: original and enhanced images with evaluation metrics.

Second row: luminance heat map of the corresponding image.

Third row: 3D surface map of the corresponding image.

As seen from the luminance heat map, each algorithm has some brightening effect on the original image. As seen from the 3D surface map, histogram equalization and logarithmic transformation enhancement have significant effects on global brightening. In this example, the histogram equalization algorithm has the highest weighted average of UCIQE and UIQM values, so the enhancement result of White Patch algorithm is chosen as the final result.

Fuzzy Enhancement

- Wiener Filter Deblurring:

Wiener filtering is a linear filtering technique used to recover a convolutionally blurred image in the presence of noise. It estimates the original image by minimizing the mean square error (MSE), combining statistical information about the signal and noise.

Equation

The frequency domain formula for Wiener filtering is as follows:

$$H_{\text{Wiener}}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + K}$$

$$F_{\text{out}}(u,v) = H_{\text{Wiener}}(u,v) \cdot G(u,v)$$

- Richardson-Lucy Deblurring:

  Richardson-Lucy algorithm is an iterative approach, based on a probabilistic model, to recover the original image by maximizing the likelihood function of the observations. It is suitable for dealing with Poisson noise and optical blur.

  The update formula for each iteration is:

$$f^{(t+1)}(x,y) = f^{(t)}(x,y) \cdot \left(\frac{g(x,y)}{(f^{(t)} * h)(x,y) + \epsilon} * h^*(x,y) \right)$$

- Guided Filter Deblurring:

  Guided filtering is an edge-preserving smoothing technique that effectively removes noise or blurring by using a guided image to direct the filtering process while preserving the edge information of the image. In deblurring, guided filtering is used to enhance the edge details, thereby improving the image clarity.

  The basic formula for guided filtering is:

$$\mathbf{I}(x) = a(x)\mathbf{G}(x) + b(x)$$

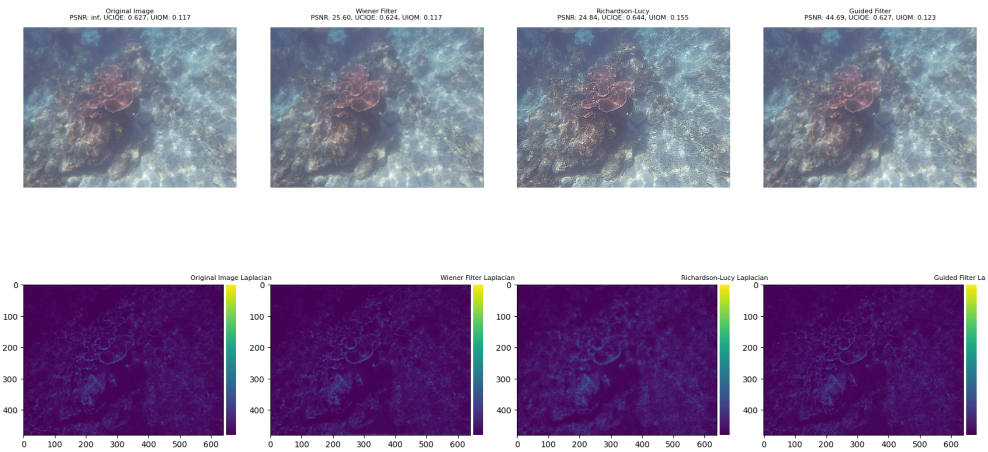

figure17:Edge detection heat map

As shown in the figure17:

First row: original and enhanced images with evaluation metrics.

Second row: Laplace transform edge detection heat map of the corresponding image

From the Laplace transform edge detection heat map, it can be seen that each algorithm has some de-blurring effect on the original image. In this example, the Richardson-Lucy deblurring algorithm has the highest weighted average of UCIQE and UIQM values, so the enhancement result of White Patch algorithm is chosen as the final result.

Calculation of metrics :

- Peak Signal-to-Noise Ratio (PSNR):
  PSNR is a metric used to measure the similarity of two images, usually used to evaluate the effect of image enhancement, compression or denoising algorithms. the higher the value of PSNR, the less correlated it is with the original image.
  Equation

$$\mathrm{PSNR} = 20 \cdot \log_{10}\left(\frac{\mathrm{MAX}_{\mathrm{pixel}}}{\sqrt{\mathrm{MSE}}}\right)$$

- UCIQE (Underwater Color Image Quality Evaluation):
  UCIQE evaluates the color quality of underwater images by considering the saturation coefficient of variation, brightness contrast and average saturation of the image.
  Formula

$$\mathrm{UCIQE} = c_1 \cdot \mathrm{SatVar} + c_2 \cdot \mathrm{Con}_L + c_3 \cdot \mathrm{AvgSat}$$

- UIQM (Underwater Image Quality Measure):
  UIQM provides a comprehensive evaluation index of underwater image quality by taking color balance (UICM), sharpness (UISM) and contrast (UIConM) into consideration, UIQM is able to effectively evaluate the overall quality of the image.
  Formula

$$\mathrm{UIQM} = c_1 \cdot \mathrm{UICM} + c_2 \cdot \mathrm{UISM} + c_3 \cdot \mathrm{UIConM}$$

  Specific indicator formula
  1) Color Balance Metrics (UICM)

$$\mathrm{UICM} = -0.0268 \cdot \sqrt{\sigma_R^2 + \sigma_Y^2} + 0.1586 \cdot \sqrt{\mu_R^2 + \mu_Y^2}$$

  2) Unclarity Indicator (UISM)

$$\text{UISM} = \frac{\text{EME}}{N}$$

3) Contrast Metrics (UIConM)

$$\text{UIConM} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\sigma_{\text{local}}(i,j)$$

Optimal algorithm selection :

We define the score by weighting UCIQE and UIQM, and judge the best algorithm based on the score. We believe that UIQM is closer to the human eye judgment standard, so we set the weighting to 30%for UCIQE and 70% for UIQM.

3.3.3 *Analysis of the Result*

File Name	Image Class	PSNR	UCIQE	UIQM
test001.png	Color Cast	29.853	0.564	0.437
test002.png	Color Cast	17.291	0.782	0.455
test003.png	Color Cast	10.256	0.602	0.443
test004.png	Color Cast	23.727	0.662	0.0.728
test005.png	Color Cast	10.600	0.638	0.450
test006.png	Color Cast	10.649	0.731	0.494
test007.png	Low Light	16.317	0.807	1.314
test008.png	Low Light	6.689	0.936	0.522
test009.png	Blur	47.339	0.572	0.506
test010.png	Color Cast	15.772	0.604	0.387
test011.png	Color Cast	18.180	0.661	0.454
test012.png	Low Light	12.723	0.901	0.410

Table 4 Enhance results

As seen in the table4:

According to the selection model proposed in the previous section, the category to which the test image belongs is judged and then the corresponding single scene enhancement model is selected according to the category. The model has some enhancement effect on all the images. Meanwhile, according to the statistical results, White patch algorithm is mostly selected for Color Cast type images. For Low Light type images, enhancedclahegamma and histogram algorithms output a comparable number of images. And the only image of Blur category is enhanced with guidedfilter algorithm.

figure19:Enhancement comparison chart

a: image before color shift enhancement b: image after color shift enhancement c: image before low light enhancement d: image after low light enhancement e: image before blur enhancement f: image after blur enhancement

As shown in the figure:

The model enhancement is obvious.

3.4 Deep Learning Enhanced Model

3.4.1 *Terms, Definitions and Symbols*

Nomenclature / Notation	Define
epoch	Number of training rounds
Lr	learning rate
Loss	loss value
Optimizer	optimizer

Table 5

3.4.2 *Modelling and Solving*

This deep learning model is based on U-Net architecture for underwater image enhancement in complex scenes.

Data preparation :

Since we are not professional researchers, we cannot directly obtain first-hand underwater image datasets. Therefore, we use C. Li's public dataset 'An Underwater Image Enhancement Benchmark Dataset and Beyond' on Github for training. The dataset contains 890 original underwater images and corresponding high quality reference images as well as 60 challenging test images.

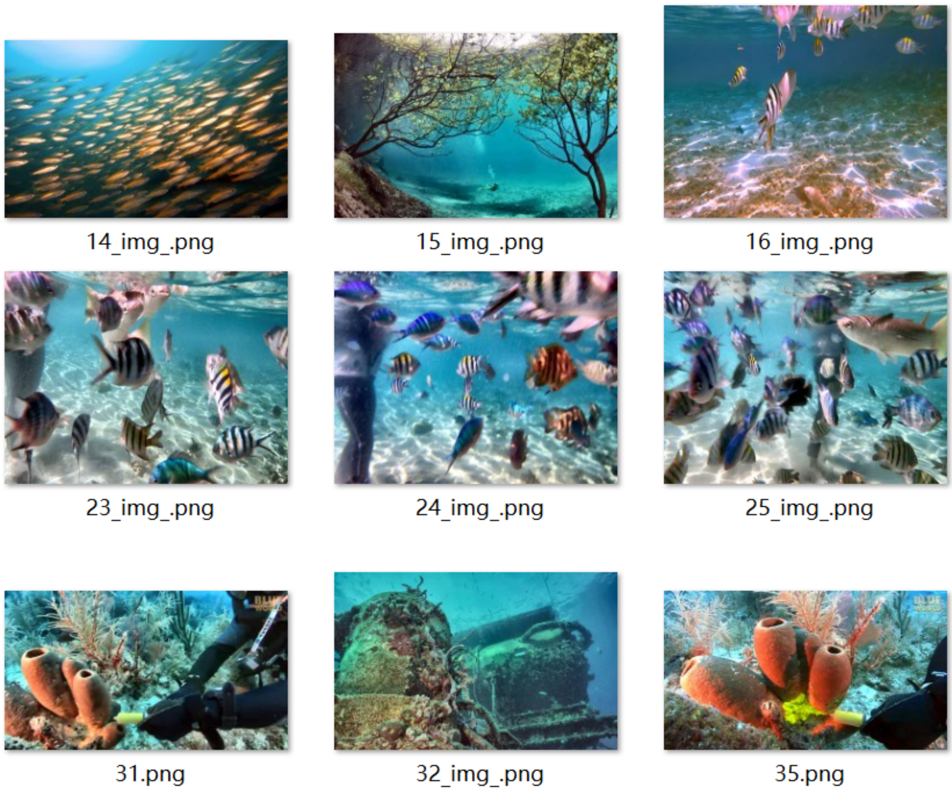

figure20:Part of the test set image

As seen in figure20:

The dataset are all real ocean clear images with high quality and training value.

Model Architecture ：

This deep learning model is based on the U-Net architecture.U-net is a neural network architecture designed primarily for image segmentation.[4] This convolutional network can use fewer training images but still produce accurate image segmentation.U-net is an encoder-decoder neural network architecture that contains the following model components:

- Encoder:

  Convolutional layer: Extracts image features, gradually increasing the number of channels (from 3 to 64 to 128).

  Activation function: Increase the nonlinearity using the ReLU activation function.

  Pooling Layer: Reduce the spatial dimension of the feature map using maximum pooling.

  Attention Module.
- Convolution Layer: Use 1x1 convolution to maintain the number of channels in the feature map.

Sigmoid Activation: Outputs weight factors to weight the feature map.

- Decoder:

  Transpose Convolutional Layer: Up-sample the feature map to restore spatial dimensions.

  Convolutional Layer: Reduce the number of channels, and finally output a 3-channel enhanced image.

Model Training Process The first step is to perform model and optimizer initialization:Firstly instantiate the U-Net model and move it to GPU.Secondly use Mean Square Error (MSE) as loss function and Adam optimizer for parameter update.

Adam optimization algorithm updates the formula:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

The second step is to perform a training loop. In this model, we set up 20 training epochs. for each epoch, the model is set to training mode and then traverses each batch in the data loader.

Finally, after the training is complete, we will save the model parameters.

Model assessment :

In this model, considering the continuity of the evaluation indexes, we use UCIQE and UIQM weighting for evaluation as in the previous section.

At the end of model training, set the model to evaluation mode and stop the gradient calculation. Then, traverse each batch in the test set. Degraded images are acquired and enhanced images are obtained by the model, and the model outputs and true clear images are converted to NumPy arrays. Then calculate PSNR, UCIQE and UIQM metrics between each pair of output images and real images and store them. Finally, the PSNR, UCIQE, and UIQM metrics values are calculated for all samples

3.4.3 *Analysis of the Result*

In the testing process, we first used the 400 graphs given in Annex 1 for testing.

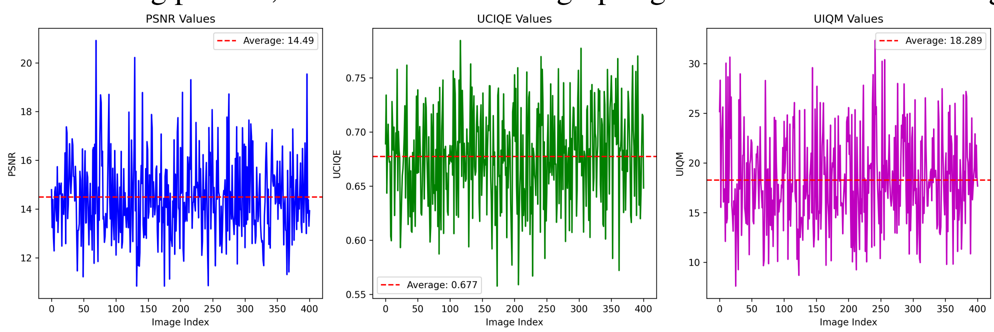

figure21:Figure indicator values in Attachment 1

As seen in figure21:

With a large number of test sets, the model performs well, and the UCIQE and UIQM metrics are both improved to some extent. This also indirectly verifies the accuracy of the model.

Finally, we use the 12 images in Appendix 2 for enhancement. The final results are obtained.

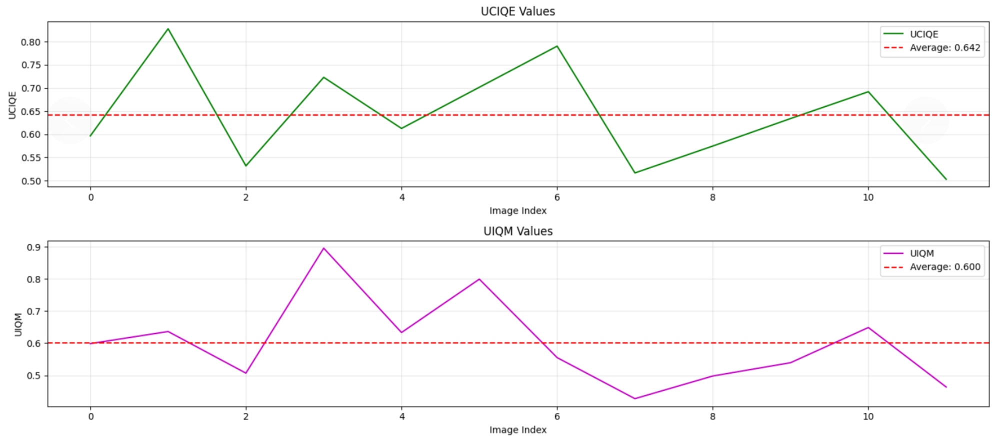

figure22:The UCIQE and UIQM values of the image in Attachment 2

As seen in the figure22:

The horizontal coordinates indicate the number of picture labels and the vertical coordinates indicate the numerical values.

For the pictures in Annex 2, the average enhancement effect of the algorithm proposed in this model is significant.

3.5 Physically Enhanced Models For Complex Scenes

3.5.1 *Modelling and Solving*

Model Building :

We mainly focus on the physical method image enhancement model under a single scene, and propose the physical method image enhancement model for complex scenes. Same as in single scene, in this model, we mainly perform image enhancement by image processing algorithms and judge the final algorithm according to the weighted average of UCIQE and UIQM, so as to output the enhanced image with the highest score.

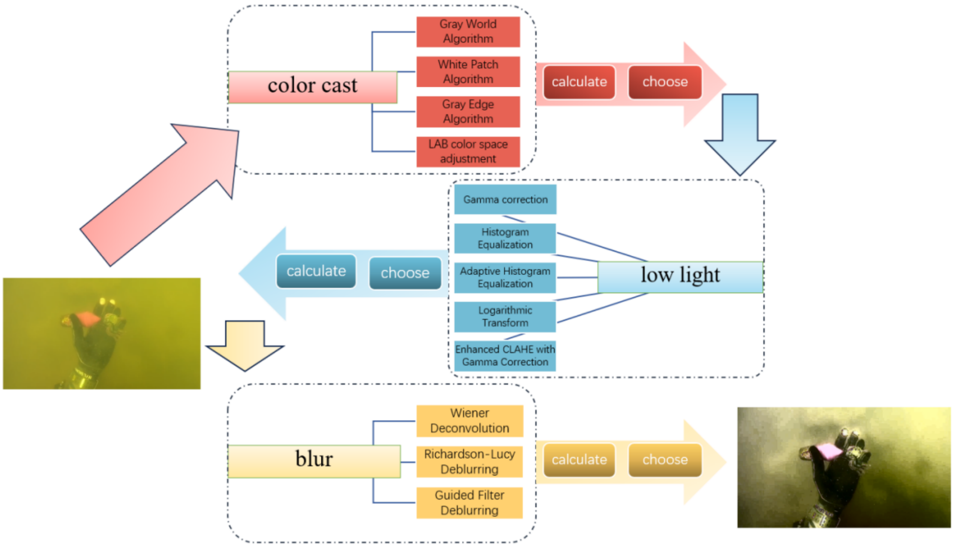

figure23:Flowchart of the physical model of a complex scene

Since the simultaneous occurrence of color shift, low brightness, and blurring is common in marine scenes, thus for most of the samples, the simultaneous enhancement of all three scene algorithms can be used for image processing.

Therefore, unlike the physical approach image enhancement model in a single scene, we remove the step of single scene screening by the classification model, and directly perform white balance enhancement, luminance enhancement, and blurring enhancement on the image in turn, and finally output the image.

Model solving :

The introduction of the algorithm of the model and its mathematical principles have been mentioned in the previous section, so we will not write too much about it here.

3.5.2 *Analysis of the Result*

The enhancement of the 12 images in Annex 2 yields the following results

As can be seen from the table6:

All the values of the image quality judgment indexes are high, and the model and its enhancement algorithm have a significant enhancement effect on the image.

File Name	PSNR	UCIQE	UIQM
test001.png	10.258	1.027	2.626
test002.png	9.868	0.896	1.864
test003.png	11.705	0.887	0.547
test004.png	11.130	0.986	2.152
test005.png	10.003	0.779	1.342
test006.png	10.560	1.015	0.696
test007.png	12.393	0.868	1.788
test008.png	6.746	0.954	0.600
test009.png	9.577	0.947	2.366
test010.png	11.757	1.004	0.533
test011.png	13.239	1.018	0.675
test012.png	11.591	0.948	0.675

Table 6 Calculate parameter values for complex scene physics models

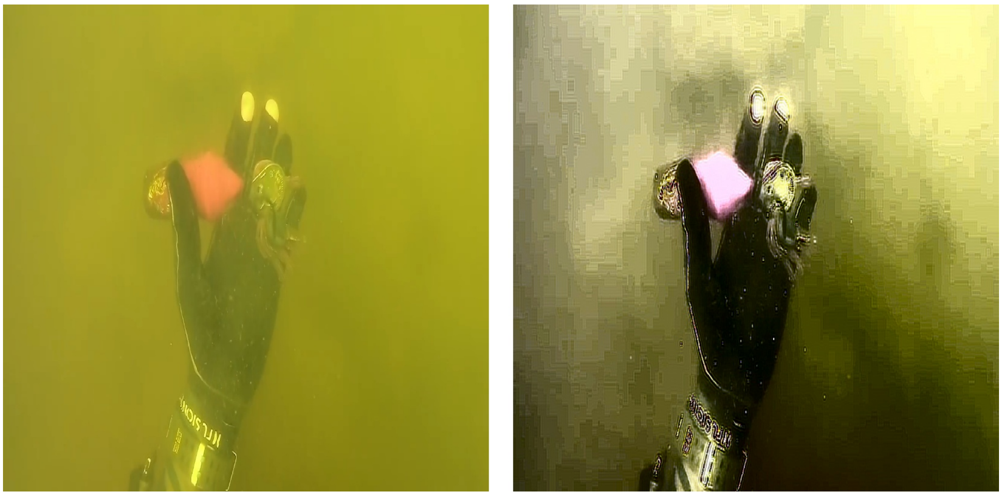

figure25:Complex scene physics model enhances the picture

The left image is the pre-enhancement image and the right image is the post-enhancement image. It can be seen from the figure:

The model has some enhancement on the image in the three indicators of white balance, brightness and sharpness, and the image observation effect is better after enhancement, which has high practical value.

3.5.3 *Comparison with Deep Learning Model - Solution to Problem 4*

Both the present model and the deep learning model are utilized in complex scenes. By comparing the UCIQE and UIQM of the output results of the two models horizontally, it is easy to find that the present model is better in terms of parameters and produces the best quality images.

Therefore, for complex scenes, we are able to get the best optimization scheme by using the physical enhancement method under complex scenes.

3.5.4 *Comparison with a single scene physical model - Solution to Problem 5*

Compare the parameters of the two model outputs side by side

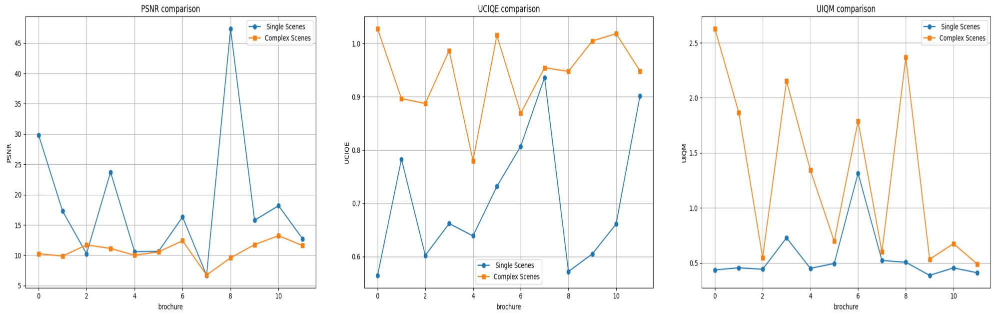

figure26:Comparison of the parameters of the two scenarios of the physical enhancement model

As shown in the figure26:

The blue line is the output result of the single scene enhancement model and the orange line is the output result of the complex scene enhancement model. In the two most critical metrics for measuring image quality, UCIQE and UIQM, all the results of the complex scene model are higher than the single scene model. However, in the PSNR metric, most of the data between the two are not much different, and a small amount of data of the single scene model has a higher value than the complex scene model.

In summary, the enhancement effect of the complex scene enhancement model is significantly higher than that of the single scene enhancement model. In the future, underwater visual enhancement technology should be developed in the direction of complex scene enhancement.

IV. Conclusions

4.1 Conclusions of the problem

4.1.1 *Underwater Image Quality Problems*

Underwater images usually suffer from severe color distortion, low contrast and blurring due to physical factors such as light attenuation, scattering and absorption. Therefore, restoring the true color, clarity and detail of underwater images is an urgent problem.

4.1.2 *Technical Challenges*

Noise and distortion in underwater images complicate image enhancement. Due to the diversity of water bodies (e.g., different depths, transparency, and water quality), different results are often encountered when applying the same enhancement method in different environments.

4.1.3 *Lack of standardized datasets*

Currently, there is a lack of standardized datasets to validate the effectiveness of algorithms in research on underwater image enhancement, which makes the results of different studies less comparable. In addition, the diversity in datasets (different water body environments and shooting conditions) adds to the complexity of the problem.

4.1.4 *Real-time issues of processing methods*

Especially for real-time underwater robotics or ocean exploration applications, image enhancement methods need to ensure low computational complexity to accommodate real-time processing.

4.1.5 *Methods Used in Our Models*

4.1.6

In classification models, normalization is an important data preprocessing method, which is used to scale the values of different features to the same range, in order to avoid the impact of large differences in values between features on the model results. Especially in weighting calculation, individual features are given different weights according to their importance, and normalization ensures that the contribution of each feature in the weighting calculation is fair. A composite score is calculated by performing a weighted sum after normalizing all features to a uniform scale. This score serves as the basis for categorization and is used to assign samples to different categories.

4.1.7 *Physical model*

Physical method enhancement model is divided into a single scene and complex scene two kinds, its core algorithm are twelve physical algorithms for picture enhancement, a single scene only use a category of enhancement algorithms, complex scene is a comprehensive use of a variety of categories of enhancement methods, and then by the weighted scores of the two indicators of the UCIQE and the UIQM, to arrive at the final algorithm and to generate the picture.

4.1.8 *Deep Learning Model*

The deep learning model is based on the U-Net architecture, an encoder-decoder neural network architecture, which is a convolutional network that can use fewer training images but still produce accurate image segmentation. This makes it suitable for use in training tasks with a small number of samples.

4.1.9 *Applications of Our Models*

Underwater visual enhancement techniques can be optimized for specific problems in a given scene, such as color correction, low-light and blur processing for different lighting conditions or water quality. However, in complex underwater environments (e.g., varying depths, turbidity, and dynamic objects), it is often difficult for a single enhancement method to cope with all the variations. Complex scenarios, on the other hand, have significant improvements with a combination of multiple classes of enhancement methods and deep learning methods that can be applied to the vast majority of scenarios:

4.1.10 *Underwater robot vision*

Our enhancement methods can provide underwater robots with clearer visual data to help them achieve autonomous navigation and target recognition in complex underwater environments. The enhanced images not only improve the robot's ability to detect objects, but also help it avoid obstacles and perform tasks effectively even in low visibility.

4.1.11 *Marine scientific research*

For oceanographers and ecologists, underwater image enhancement technology can provide clearer images of marine life, coral reefs and other underwater landscapes. This technology can improve the accuracy of ecological monitoring, especially in deep-sea exploration and ecological assessment, helping scientists to observe and analyze underwater ecosystems more accurately.

4.1.12 *Underwater Remote Sensing*

In underwater remote sensing missions, enhancing the quality of images is critical to improving the accuracy of identifying features in the water column. For example, image enhancement techniques can help accurately identify changes in water quality, pollutant distribution, and other environmental features that are important for underwater monitoring and environmental protection.

V. Future Work

5.1 Real-time improvement of enhancement algorithms

Although current underwater image enhancement methods can significantly improve image quality, they still face the problem of slow computation in some application scenarios, especially in real-time underwater robotics and ocean exploration. Therefore, future research can focus on improving the real-time performance of image enhancement algorithms, such as reducing computational complexity through lightweight deep learning networks to realize real-time processing.

5.2 Cross-environmental adaptability

Although the method proposed in this study achieves good enhancement results in specific environments, the image enhancement effect in different waters, depths, transparency and other conditions still needs to be further improved. Future work can focus on cross-environmental adaptation and explore how to design more pervasive models that can automatically adjust the parameters to adapt to different underwater environments, and even maintain stable effects under dynamically changing environments. Adaptive modeling based on environment perception may become an important direction.

5.3 Multimodal data fusion

In addition to image enhancement, future research can also explore the use of other sensor data (e.g., sonar data, LIDAR data, etc.) to further enhance target recognition and image reconstruction in underwater environments. Through multimodal data fusion, missing information in underwater images can be supplemented to achieve more comprehensive and accurate underwater image enhancement. For example, combining sonar images and visual images may be able to overcome the problem of incomplete underwater visual information and improve the overall image quality and accuracy.

5.4 Diversity and Standardization of Data Sets

Although the existing datasets are representative in some aspects, due to the diversity of underwater environments (e.g., different depths, turbidity, lighting conditions, etc.), the existing datasets still cannot cover all application scenarios. Therefore, future work could focus on creating richer and more diverse underwater image datasets to better validate the generalization ability of different algorithms. In addition, the development of standardized datasets and evaluation metrics will help to promote the advancement of underwater image enhancement techniques and enable fairer and more consistent comparisons between different methods.

5.5 Combination of underwater image and video enhancement

Most of the current research on underwater image enhancement focuses on static images, however, enhancement of underwater video, especially in dynamic environments, is still a more complex subject. In the future, the quality of underwater video can be improved by video enhancement algorithms by combining time series information. Especially when dealing with underwater motion blur, low-light and color distortion, the use of spatio-temporal consistency may be able to obtain more stable and efficient results.

VI. References

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